

Modeling Adaptive Grasping in a Cable-Driven Origami Dexterous Hand

From posture synergies and origami tendon routing to SDAS/FMAs-based MuJoCo contact simulation

Problem Statement

Human hands have many joints, but grasping is often coordinated by a few low-dimensional postural synergies. The `synergy_hand_s1a` project studies this idea in an origami-inspired dexterous hand whose folds, compliant crosses, holes, actuators, and routed tendons are all represented in a design file.

The modeling challenge is that the hand is not a fully actuated rigid linkage. Its origami folds and tendons create a differential-like underactuated mechanism: when one part is constrained by an object, tendon motion and joint compliance can redistribute motion to other joints. A two-ended closed-loop tendon is especially important because pulling from side A and side B creates different routed tendon paths.

$$q \approx S^{(k)} \sigma^{(k)}, \quad k \ll n$$

$$q_{ref}(t) = B_R \sigma(t)$$

$$z = Rq, \quad R = \text{constant} \implies R \rightarrow R(S)$$

Origami Hand CAD — ohd_6 设计视图

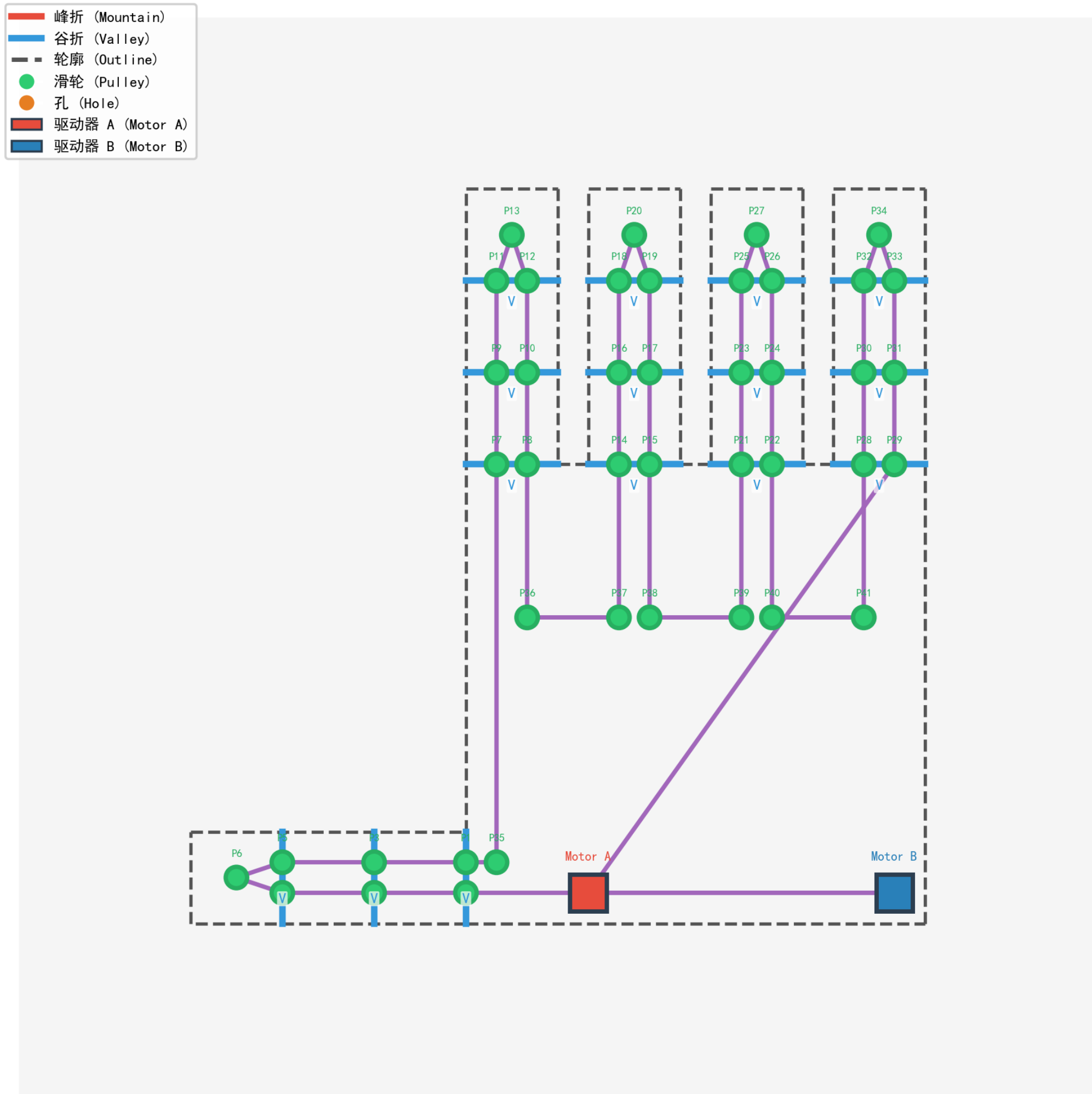


Fig. 1. The project represents an origami hand with faces, crosses, joints, holes, actuators, and tendon routes.

Literature Review

Rigid synergy compresses high-dimensional hand posture into a low-dimensional command. Soft synergy treats that command as a compliant reference rather than a rigid final shape. Adaptive synergy connects the idea to hardware through tendon or differential transmissions. The SDAS/FMAs model proposed here extends adaptive synergy to a two-ended closed-loop tendon.

Rigid: $q = S\sigma$

Soft: $q_p = S\sigma, \quad J^T f_e = K(q_p - q)$

Adaptive: $Rq = z, \quad q = E^{-1}R^T(R E^{-1}R^T)^{-1}z$

SDAS/FMAs: $R_A = A_A^T \bar{R}, \quad R_B = A_B^T \bar{R}$

$$\tau = R_A^T F_A + R_B^T F_B$$

The key gap is the fixed-transmission assumption. A two-ended tendon needs two one-sided transmissions because the tendon distribution depends on the pulling direction.

Methodology

The project has three connected layers. First, the design and geometric simulation layer parses `.ohd/DXF` data, builds topology, exports visual geometry, and maps low-dimensional commands to reference joint motion. Second, SDAS/FMAs derives a tendon-based synergy from two-ended closed-loop routing. Third, MuJoCo solves the resulting dynamics and contact constraints.

$$\delta l = \bar{R} \delta q$$

$$T_j^{(A)} = \alpha_j^{(A)} F_A, \quad T_j^{(B)} = \alpha_j^{(B)} F_B$$

$$R_A = A_A^T \bar{R}, \quad R_B = A_B^T \bar{R}$$

$$q = S_A x_A + S_B x_B + C J^T f_e$$

$$\sigma = \frac{x_A + x_B}{2}, \quad \sigma_f = \frac{x_A - x_B}{2}$$

$$q = (S_A + S_B)\sigma + (S_A - S_B)\sigma_f + C J^T f_e$$

$$M(q)\ddot{q} + h(q, \dot{q}) = \tau_{SDAS}(t) + J_c(q)^T \lambda$$

图 3-2 Capstan 指数衰减导致的 R_A 与 R_B 分布

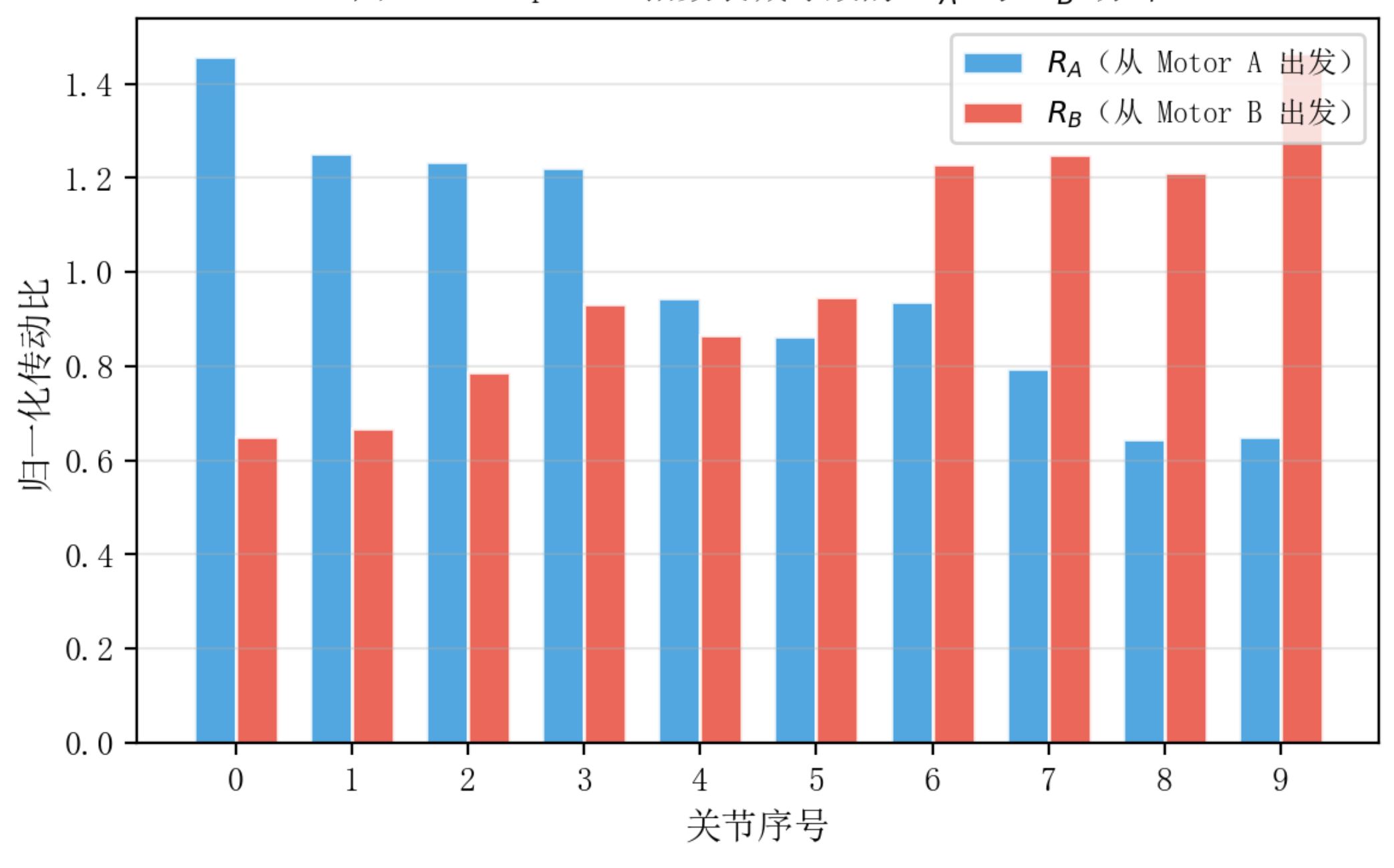


Fig. 2. SDAS/FMAs separates the A-side and B-side tendon transmissions.

Experimental Results

The results are selected to show the full project chain: design representation, reference motion, numerical state, and contact grasping. A free-space step input compares the geometric reference and the MuJoCo state. Contact scenes then show how SDAS/FMAs actuation behaves under object constraints.

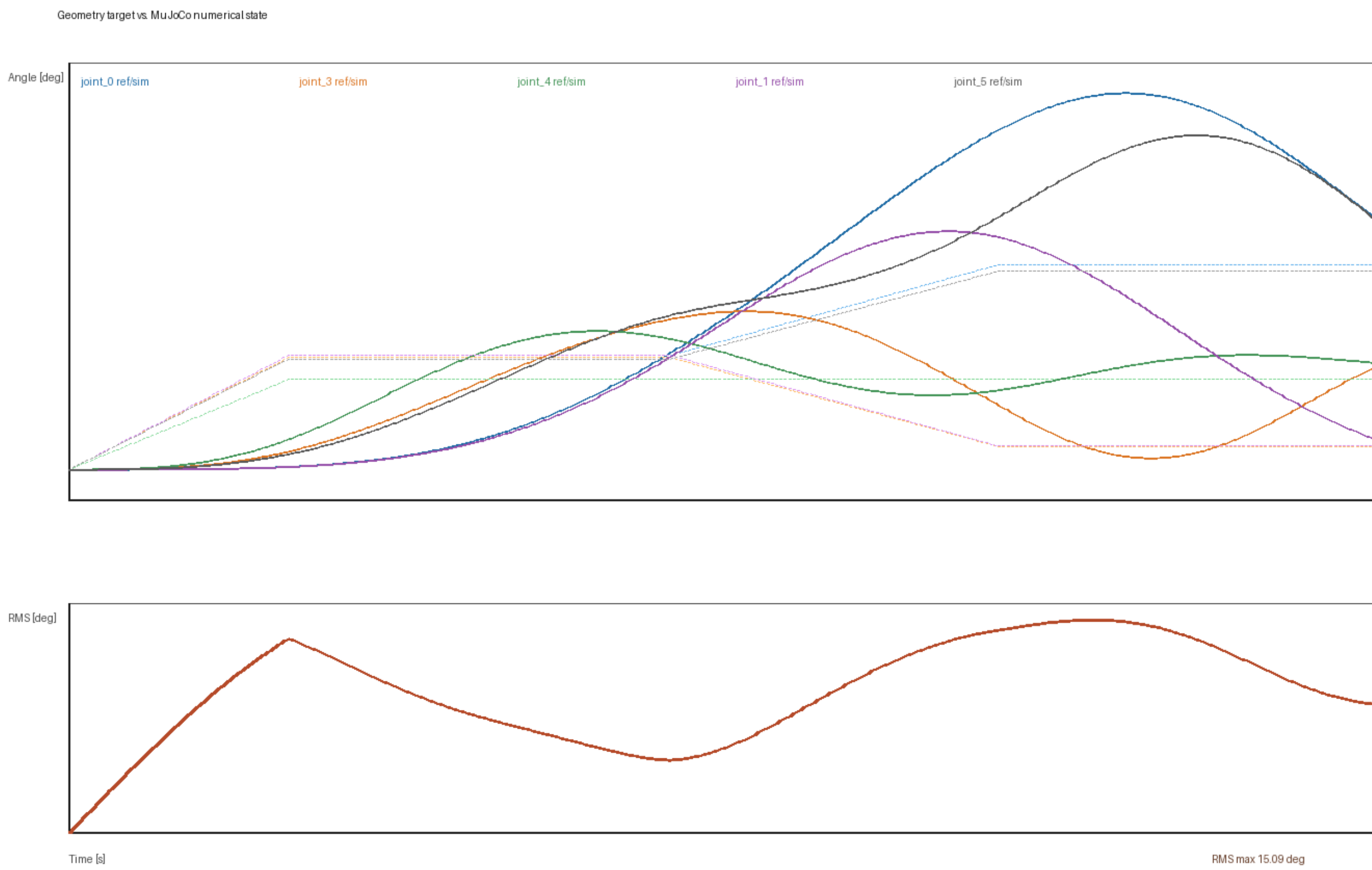


Fig. 3. Geometry/SDAS reference versus MuJoCo numerical state.

Representative Contact Demos

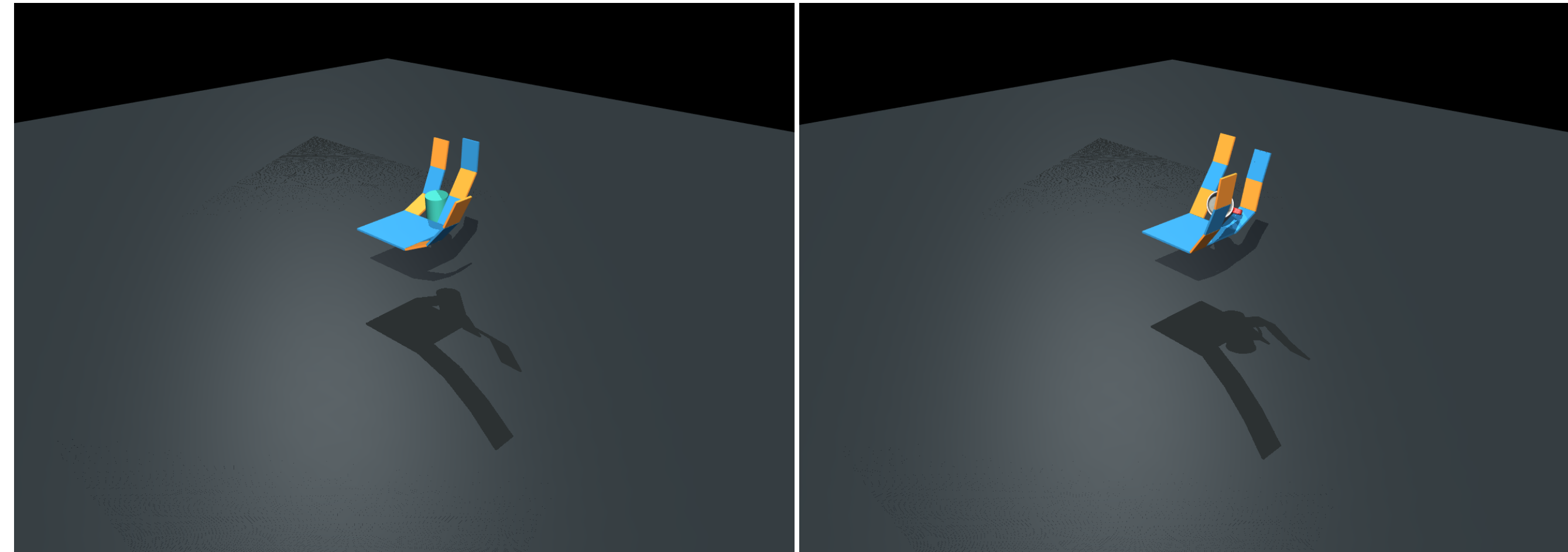


Fig. 4. Cylinder and scanned-mug grasping demos.

Demo	Steps	Contact steps	Max contacts	RMS
Free step	600	0	0	0.1848
Cylinder	700	158	5	0.1816
Scanned mug	700	209	7	0.1821

$$F_{\text{contact, total}}(t) = \sum_{i=1}^{n_c(t)} \|J_{c,i}(t)\|$$

Conclusion

The contribution is a full modeling chain: origami hand design, geometric synergy simulation, SDAS/FMAs tendon transmission, and MuJoCo contact-aware numerical grasping. The project does not only render a final posture; it explains how the designed tendon routing produces a low-dimensional actuation model and how that model behaves under numerical dynamics and object contact.

`.ohd design` $\rightarrow$ `qref` $\rightarrow$ τ_{SDAS} $\rightarrow$ MuJoCo contact simulation

References: Santello et al. 1998; Bicchi et al. 2011; Catalano et al. 2014; Todorov et al. 2012.